

THE RING OF CYCLOTOMIC INTEGERS

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1. INTRODUCTION

For an integer $n \in \mathbb{N}$, let $\mathbb{Q}(\zeta_n)$ denote the n th cyclotomic field. In this note, we prove that the ring of integers of $\mathbb{Q}(\zeta_n)$ satisfies the natural equality

$$(1) \quad \mathcal{O}_{\mathbb{Q}(\zeta_n)} = \mathbb{Z}[\zeta_n]$$

The proof proceeds in two steps. First, we show that (1) holds when $n = p^r$ is a prime power by passing to the p -adic numbers $\mathbb{Q}_p$. Then, for a general integer $n = p_1^{r_1} \cdots p_n^{r_n}$, we prove (1) by combining multiplicatively across the prime powers $p_i^{r_i}$.¹

By using the p -adic numbers, we avoid many of the technical details found in standard proofs of (1) (see e.g. [3, Ch. 1, §10, Lem. 1, Prop. 2]). This nicely illustrates a general philosophy in algebraic number theory: to show a statement is true over $\mathbb{Q}$, it is often preferable to “pass to local fields” and prove it locally, then glue the pieces together to get the global picture.

2. THE CASE $n = p^r$

In this section, we will prove (1) when $n = p^r$ is a prime power. It is here that we use $\mathbb{Q}_p$ to significantly simplify matters. First, we will need two technical lemmas.

Lemma 2.1. *Let $A \subseteq B$ be two finitely generated, torsion-free abelian groups of rank $n \in \mathbb{N}$. Then for each prime p ,*

$$[B \otimes_{\mathbb{Z}} \mathbb{Z}_p : A \otimes_{\mathbb{Z}} \mathbb{Z}_p] = p^{v_p([B:A])},$$

where left-hand side is the cardinality of the $\mathbb{Z}_p$ -module $(B \otimes_{\mathbb{Z}} \mathbb{Z}_p)/(A \otimes_{\mathbb{Z}} \mathbb{Z}_p)$, and v_p denotes the p -adic valuation.

Proof. Choose $\mathbb{Z}$ -bases so that the inclusion $A \hookrightarrow B$ is represented by a matrix $T \in M_n(\mathbb{Z})$ with determinant $d \neq 0$, so $[B : A] = |d|$. After tensoring by $\mathbb{Z}_p$, the inclusion $A \otimes_{\mathbb{Z}} \mathbb{Z}_p \hookrightarrow B \otimes_{\mathbb{Z}} \mathbb{Z}_p$ is still represented by T (now viewed as a matrix over $\mathbb{Z}_p$), thus over the PID $\mathbb{Z}_p$, we can put T into Smith normal form as follows:

$$U T V = \text{diag}(p^{a_1}, \dots, p^{a_n}), \quad U, V \in \text{GL}_n(\mathbb{Z}_p), \quad a_i \geq 0.$$

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¹The idea for this proof originated from a comment [1] by Keith Conrad on MathOverflow.

The cokernel of $T : A \otimes \mathbb{Z}_p \rightarrow B \otimes \mathbb{Z}_p$ is thus

$$(B \otimes \mathbb{Z}_p)/(A \otimes \mathbb{Z}_p) \cong \bigoplus_{i=1}^n \mathbb{Z}_p/(p^{a_i}) \cong \bigoplus_{i=1}^n \mathbb{Z}/p^{a_i}\mathbb{Z},$$

so

$$[B \otimes \mathbb{Z}_p : A \otimes \mathbb{Z}_p] = p^{a_1 + \dots + a_n} = p^{v_p(\det T)} = p^{v_p([B:A])}.$$

□

Lemma 2.2. *For a prime power p^r , $\mathbb{Z}_p[\zeta_{p^r}]$ is a discrete valuation ring.*

Proof. Using

$$\Phi_{p^r}(x) = \prod_{\substack{1 \leq a \leq p^r \\ (a, p^r)=1}} (x - \zeta_{p^r}^a),$$

we obtain

$$\Phi_{p^r}(1) = p = \prod_{\substack{1 \leq a \leq p^r \\ (a, p^r)=1}} (1 - \zeta_{p^r}^a).$$

The factors $(1 - \zeta_{p^r}^a)$ are Galois conjugates of $1 - \zeta_{p^r}$, so they all generate the same prime ideal $(1 - \zeta_{p^r})$ above p . That is, (p) is a power of $(1 - \zeta_{p^r})$. But every maximal ideal in $\mathbb{Z}_p[\zeta_{p^r}]$ contains (p) , hence $(1 - \zeta_{p^r})$ is the unique maximal ideal in $\mathbb{Z}_p[\zeta_{p^r}]$. In particular, $\mathbb{Z}_p[\zeta_{p^r}]$ is a local ring. But $\mathbb{Z}_p[\zeta_{p^r}]$ is also finite over the DVR $\mathbb{Z}_p$, so in particular $\mathbb{Z}_p[\zeta_{p^r}]$ is a local Dedekind domain. Since every local Dedekind domain is a DVR, we have the result. □

Having established these two lemmas, the proof becomes relatively straightforward.

Theorem 2.3. *Let $K := \mathbb{Q}(\zeta_{p^r})$ for a prime power p^r . Then $\mathcal{O}_K = \mathbb{Z}[\zeta_{p^r}]$.*

Proof. Since the inclusion $\mathbb{Z}[\zeta_{p^r}] \subseteq \mathcal{O}_K$ is clear, it suffices to show that

$$[\mathcal{O}_K : \mathbb{Z}[\zeta_{p^r}]] = 1,$$

where $\mathcal{O}_K$ and $\mathbb{Z}[\zeta_{p^r}]$ are viewed as free $\mathbb{Z}$ -modules. By [2, Ch. 1, Ex. 27(c)], we have

$$(2) \quad \text{disc}(\mathbb{Z}[\zeta_{p^r}]) = \text{disc}(\mathcal{O}_K) \cdot [\mathcal{O}_K : \mathbb{Z}[\zeta_{p^r}]]^2.$$

Since it is known that $\text{disc}(\mathbb{Z}[\zeta_{p^r}])$ is a signed power of p , (2) implies that $[\mathcal{O}_K : \mathbb{Z}[\zeta_{p^r}]]$ is a power of p . That is,

$$(3) \quad [\mathcal{O}_K : \mathbb{Z}[\zeta_{p^r}]] = p^{v_p([\mathcal{O}_K : \mathbb{Z}[\zeta_{p^r}]])},$$

Now, tensoring by $\mathbb{Z}_p$, we have

$$\mathbb{Z}[\zeta_{p^r}] \otimes_{\mathbb{Z}} \mathbb{Z}_p \cong \mathbb{Z}_p[\zeta_{p^r}], \quad \mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p \cong \mathcal{O}_{K_p},$$

where $K_p := \mathbb{Q}_p(\zeta_{p^r})$. By Lemma 2.1,

$$[\mathcal{O}_{K_p} : \mathbb{Z}_p[\zeta_{p^r}]] = p^{v_p([\mathcal{O}_K : \mathbb{Z}[\zeta_{p^r}]])},$$

thus by (3),

$$[\mathcal{O}_K : \mathbb{Z}[\zeta_{p^r}]] = [\mathcal{O}_{K_p} : \mathbb{Z}_p[\zeta_{p^r}]].$$

Therefore, it suffices to show that $\mathcal{O}_{K_p} = \mathbb{Z}_p[\zeta_{p^r}]$. By 2.2, $\mathbb{Z}_p[\zeta_{p^r}]$ is a discrete valuation ring in K_p , hence it is integrally closed in K_p . Since $\mathbb{Z}_p \subseteq \mathbb{Z}_p[\zeta_{p^r}]$, we have that $\mathbb{Z}_p[\zeta_{p^r}]$ must be the integral closure of $\mathbb{Z}_p$ in K_p . But $\mathcal{O}_{K_p}$ is the unique subring of K_p with this property, thus $\mathcal{O}_{K_p} = \mathbb{Z}_p[\zeta_{p^r}]$ as desired. $\square$

3. EXTENSION TO GENERAL CYCLOTOMIC FIELDS

Having sorted out the prime power case, the general equality follows relatively quickly.

Theorem 3.1. *For any integer $n \geq 1$, let $K_n := \mathbb{Q}(\zeta_n)$. Then,*

$$\mathcal{O}_{K_n} = \mathbb{Z}[\zeta_n].$$

Proof. We proceed by induction on n . First, suppose n is not a prime power. Then, we can write $n = ab$ for coprime integers $a, b > 1$. We claim that K_n is the compositum of fields $K_n = K_a K_b$, where $K_m := \mathbb{Q}(\zeta_m)$. First, since $\zeta_n^b = \zeta_a$ and $\zeta_n^a = \zeta_b$, we have $\zeta_a, \zeta_b \in \mathbb{Q}(\zeta_n)$, so $K_a K_b \subseteq K_n$. Conversely, since $\gcd(a, b) = 1$, there exist integers r, s with $ar + bs = 1$ by Bézout's identity. Thus

$$\zeta_n = \zeta_n^{ar+bs} = (\zeta_n^a)^r (\zeta_n^b)^s = \zeta_b^r \zeta_a^s,$$

so $\zeta_n \in K_a K_b$. Therefore $K_n \subseteq K_a K_b$, hence $K_n = K_a K_b$ as claimed. It follows that

$$[K_n : \mathbb{Q}] = \varphi(n) = \varphi(a)\varphi(b) = [K_a : \mathbb{Q}][K_b : \mathbb{Q}].$$

Now, by the inductive hypothesis, $\mathcal{O}_{K_a} = \mathbb{Z}[\zeta_a]$ and $\mathcal{O}_{K_b} = \mathbb{Z}[\zeta_b]$. Moreover, since $\gcd(a, b) = 1$ and the discriminant of $\mathbb{Q}(\zeta_m)$ is only divisible by primes dividing m , we have

$$\gcd(\text{disc}(K_a), \text{disc}(K_b)) = 1.$$

Therefore, by [2, Ch. 2, §7, Cor. 1],

$$\mathcal{O}_{K_n} = \mathcal{O}_{K_a} \mathcal{O}_{K_b} = \mathbb{Z}[\zeta_a] \mathbb{Z}[\zeta_b] = \mathbb{Z}[\zeta_n]$$

as desired. $\square$

REFERENCES

- [1] K. Conrad, Comment on “Quick proof of the fact that the ring of integers of $\mathbb{Q}(\zeta_n)$ is $\mathbb{Z}[\zeta_n]$ ”, *MathOverflow*, 2010, <https://mathoverflow.net/questions/17289/quick-proof-of-the-fact-that-the-ring-of-integers-of-mathbb-q-zeta-n-is-mathbb-z-zeta-n/17329>. 1
- [2] D. A. Marcus, *Number Fields*, Universitext, Springer-Verlag, New York, 1977. 2, 3
- [3] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften, Vol. 322, Springer-Verlag, Berlin, 1999. 1